

Satellite DNA sequences and reproductive isolation in the *Drosophila willistoni* group¹

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(1980)

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Summary. Satellite DNAs of the *Drosophila willistoni* group have been analyzed by Hae III endonuclease digestion. Differences were observed between the 2 species *D. willistoni* and *D. paulistorum* and also between semispecies of *D. paulistorum* and, to a lesser extent, between subspecies of *D. willistoni*. Differences did not appear to be consistently greater between strains producing sterile hybrid males than between those producing fertile hybrids.

Satellite DNAs are highly repeated sequences mainly located in heterochromatin but also in telomeric sites of eukaryotic chromosomes. The different hypotheses proposed about their function include 1. that they hinder recombination in the chromosome sections where they are located² and 2. that they play a role in speciation by hindering the pairing of homologous chromosomes during meiosis in interspecific hybrids³⁻⁵. The present work tests this 2nd hypothesis by studying highly-repeated satellite sequences

in populations at different stages of evolutionary divergence.

The *Drosophila willistoni* group of species provides ideal material for the test, because the main stages of the speciation process can be identified within the group⁶. We have examined the satellite DNA sequences in 8 different strains: 1. *D. willistoni willistoni* from Santa Marta (Colombia); 2. *D. w. willistoni* from Tame (Colombia); 3. *D. willistoni quechua* from Lima (Peru); 4. *D. w. 'quechua'* from

Belize; 5. *D. paulistorum* semispecies Amazonian from Caicara (Venezuela); 6. *D. paulistorum* Amazonian from Tame; 7. *D. paulistorum* semispecies Andean-Brazilian from Caicara; and 8. *D. paulistorum* Andean-Brazilian from Mirassol (Brazil). 4 levels of evolutionary divergence are represented in these strains: a) local populations; b) subspecies - *D. w. willistoni* and *D. w. quechua* - which produce sterile male hybrids but exhibit no sexual isolation and thus represent the first stage of speciation⁶; c) semispecies or incipient species - Amazonian and Andean-Brazilian of *D. paulistorum* - which produce sterile male hybrids and exhibit some sexual isolation and thus represent the second stage of speciation; d) fully reproductively isolated but closely related species - *D. willistoni* and *D. paulistorum*.

DNA was extracted from purified nuclei of 7 to 8 g of each sex of each strain using whole bodies of adult individuals, essentially according to Cordeiro-Stone and Lee⁷. The DNA was finally purified by preparative CsCl gradient centrifugation. DNA molecular weight was determined by band sedimentation in a Spinco model E ultracentrifuge according to the method of Studier⁸.

Satellite DNA bands could not be isolated from the main band DNA of the different *Drosophila* species by the conventional methods of fractionation of total native DNA in CsCl, Hg²⁺-Cs₂SO₄ and Ag⁺-Cs₂SO₄ preparative gradients. Therefore highly repeated satellite DNAs were isolated by DNA denaturation and reannealing to a Cot of 2 or 10 and subsequent fractionation on a preparative Ag⁺-Cs₂SO₄ gradient. With this technique and using a Cot of 10, as previously shown for human and mouse DNA^{9,10}, unique, middle repeated intermediate DNA, and satellite highly repeated DNAs can be separated as distinct peaks because they bind different amounts of Ag⁺ according to the extent of their double-strandedness. At a Cot of 2, satellite DNA can be separated from unique and intermediate DNA. The satellite DNA appears to be lighter in Ag⁺-Cs₂SO₄ because, having completely reannealed, it binds less Ag⁺ than the unique and intermediate DNA (figure 1).

This technique offers the advantage of separating all the satellite sequences, since no satellite DNA remains in the middle repeated and unique DNA peaks as long as the DNA size is sufficiently smaller than the presumed length of satellite DNA blocks in the genome. For this reason, *Drosophila* DNA of a size of 6000-7000 nucleotide pairs was used. That the satellite DNA obtained by this technique is in the reannealed, rather than native, state is not a disadvantage with respect to restriction enzyme digestion,

since satellite DNA reacquires its full double-strandedness after reannealing to a Cot of 2 or higher, as judged from melting curves of reannealed DNA and increase in CsCl density on denaturation.

Hae III restriction endonuclease digestion was carried out on unique, middle repeated, and satellite DNAs of male and female individuals of each of the 8 *Drosophila* strains. Only a smear with no discrete bands was observed in the agarose gel electrophoresis pattern of DNA fragments after Hae III digestion with unique and middle repeated DNA. This confirms that no satellite sequence had remained bound to unique and middle repeated DNA. Agarose gel electrophoresis patterns of satellite DNA fragments of 4 *Drosophila* strains after Hae III endonuclease digestion are shown in figure 2. The relative amount in the fragments was estimated from the area under the peak corresponding to each fragment on microdensitometer tracings.

In *D. willistoni* the pattern of digestion manifests 11 fragments. The 2 major peaks have a size of 600,000 daltons and 410,000 daltons. There are no consistent differences between the strains or subspecies of *D. willistoni* in the amount of these 2 fragments.

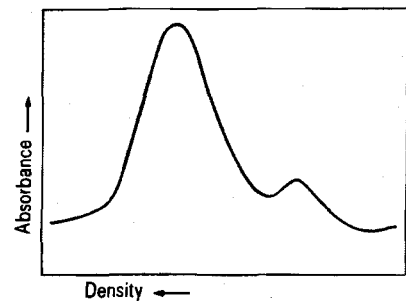


Fig. 1. Microdensitometer tracing of total nuclear DNA from female *Drosophila paulistorum* Andean-Brazilian (Caicara). The DNA was denatured and reannealed to Cot 2, then centrifuged in an analytical Ag⁺-Cs₂SO₄ equilibrium density gradient at a DNA concentration of 35 µg/ml and an Ag⁺/DNA-P molar ratio of 0.2.

Summary of the differences between *Drosophila* strains in the relative amounts of Hae III endonuclease-digested satellite DNA fragments

Strain	Fragment size in thousands of daltons					
	1200	900	760	550	510	450
<i>D. w. willistoni</i> Santa Marta	-	-	+	4%	-	7-8%
<i>D. w. willistoni</i> Tame	-	-	+	<1%	-	1-2%
<i>D. w. quechua</i> Lima	-	-	+	4%	-	7-8%
<i>D. w. quechua</i> Belize	-	-	+	<1%	-	7-8%
<i>D. paulistorum</i> Amazonian Caicara	+	-	-	10%	5%	1%
<i>D. paulistorum</i> Amazonian Tame	+	+	-	15%	1%	1%
<i>D. paulistorum</i> Andean-Brazilian Caicara	+	-	-	15%	1%	13%
<i>D. paulistorum</i> Andean-Brazilian Mirassol	+	-	-	10%	5%	1%

+, present; -, absent.

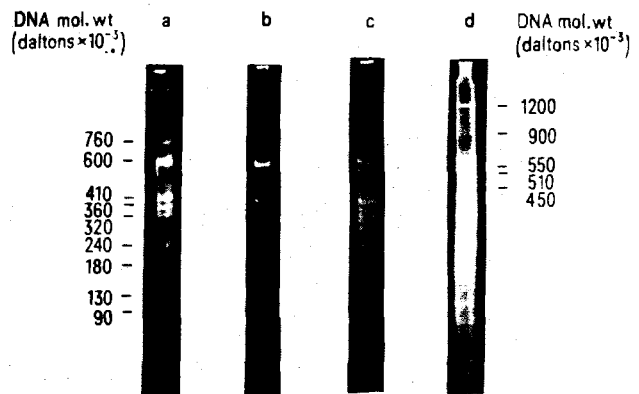


Fig. 2. Hae III restriction endonuclease digestion of satellite DNA (1.5-2 µg) from a) *Drosophila willistoni willistoni* Tame ♀; b) *D. w. quechua* Belize ♂; c) *D. paulistorum* Amazonian Caicara ♀; and d) *D. paulistorum* Amazonian Tame ♂. DNA was digested by Hae III endonuclease and run on 2.5% agarose gel electrophoresis, as reported by Southern¹¹. Fragments were sized by comparison with the mobilities of the Hae III restriction digests of lambda phage and SV40 DNA¹².

However, 2 other fragments with 550,000 and 450,000 daltons mol.wt vary considerably in relative amount in strains and subspecies. The 550,000-daltons fragment accounts for 1% or less of the total satellite DNA in *D. w. willistoni* Tame and in *D. w. quechua* Belize, whereas it accounts for about 4% of the total satellite in *D. w. willistoni* Santa Marta and in *D. w. quechua* Lima. The 450,000-daltons fragment accounts for a relative amount of 7-8% in *D. w. willistoni* Santa Marta and in the 2 strains of *D. w. quechua*, while it accounts for only 1-2% in *D. w. willistoni* Tame.

The 11 fragments observed in *D. willistoni* are present with the same mol.wts in the *D. paulistorum* strains, with the exception of the largest fragment which has a mol.wt of 1,200,000 daltons in *D. paulistorum*, instead of 760,000 daltons in *D. willistoni*. Besides this difference, the *D. paulistorum* Amazonian from Tame has a fragment with a mol.wt of 900,000 daltons that is absent in the other *D. paulistorum* strains.

The 550,000 daltons fragment is present in *D. paulistorum* satellite DNA in a relative amount of about 10% in the Amazonian semispecies from Caicara and Andean-Brazilian semispecies from Mirassol, but of 15% in the other 2 strains; these amounts are all consistently higher than those observed in *D. willistoni*. Furthermore an additional fragment of 510,000 daltons is present in *D. paulistorum* Amazonian Caicara and Andean-Brazilian Mirassol in a relative amount of 5%, and in a relative amount of 1% in the other 2 strains of *D. paulistorum*. The relative amount of the 450,000-daltons fragment, which is variable in *D. willistoni*, varies in *D. paulistorum* as well; it accounts in Andean-Brazilian Caicara for 13%, and for about 1% of the total satellite in the other 3 strains. Moreover, the 410,000-

daltons fragment is present in slightly different amounts in the *D. paulistorum* strains (see table for a summary).

In conclusion, even though the similarity in satellite-DNA digestion patterns in the strains studied reflects their close phylogenetic relationship, substantial differences exist between the 2 species, *D. willistoni* and *D. paulistorum*. Moreover, clear differences also exist between the strains of *D. paulistorum* and lesser differences between the strains of *D. willistoni*. However, strains of different subspecies or semispecies, which produce sterile hybrid males, are not consistently more different than strains of the same subspecies or semispecies, which produce fertile males and females. Hence, the sterility of the hybrid males cannot simply be explained in terms of the observed satellite DNA differences.

- 1 This work was supported by grants 78.01852.65 from the CNR (Rome) and GM 22221 from the PHS (USA).
- 2 M. Yamamoto and G.L.G. Miklos, *Chromosoma* 66, 71 (1978).
- 3 P.M.B. Walker, *Nature* 229, 306 (1971).
- 4 G. Corneo, *Experientia* 34, 1141 (1978).
- 5 G. Corneo, *Evol. Theory* 1, 261 (1976).
- 6 F.J. Ayala, M.L. Tracey, D. Hedgecock and R.C. Richmond, *Evolution* 28, 576 (1975).
- 7 M. Cordeiro-Stone and S.C. Lee, *J. mol. Biol.* 104, 1 (1976).
- 8 F.W. Studier, *J. mol. Biol.* 11, 373 (1965).
- 9 E. Ginelli and G. Corneo, *Chromosoma* 56, 55 (1976).
- 10 E. Ginelli, R. Di Lernia and G. Corneo, *Chromosoma* 61, 215 (1977).
- 11 E.M. Southern, *J. mol. Biol.* 94, 51 (1975).
- 12 P. Lebowitz, W. Siegel and J. Sklar, *J. molec. Biol.* 88, 105 (1974).